# Assignment 01

## 1. Flowchart

When the input is “a = 3, b = 7, c = 2”, the output is “No Answer”.
When the input is “a = 9, b = 8, c = 5”, the output is “9 8 5”.

## 2. Matrix multiplication

### 2.1

The matrix M1 is:
*[[2, 35, 16, 40, 20, 32, 2, 9, 27, 22],*
*[14, 14, 38, 23, 44, 49, 39, 8, 22, 37],*
*[41, 34, 41, 48, 6, 17, 39, 25, 31, 28],*
*[26, 28, 11, 40, 21, 7, 36, 37, 31, 2],*
*[32, 9, 16, 28, 25, 46, 22, 19, 44, 33]]*
The matrix M2 is:
*[[20, 10, 26, 31, 44],*
*[0, 3, 24, 19, 5],*
*[48, 24, 38, 12, 19],*
*[2, 2, 30, 32, 38],*
*[14, 43, 20, 5, 36],*
*[34, 26, 13, 40, 38],*
*[45, 4, 24, 28, 37],*
*[21, 2, 14, 8, 46],*
*[20, 31, 11, 18, 18],*
*[33, 24, 13, 3, 9]]*

### 2.2

According to the random matrices M1 and M2 generated in section 2.1, the Matrix_multip function can be used to obtain:
*[[50, 65, 40, 48, 42],*
*[47, 62, 37, 45, 39],*
*[39, 54, 29, 37, 31],*
*[77, 92, 67, 75, 69],*
*[55, 70, 45, 53, 47]]*

## 3. Pascal triangle

The detailed results are provided in the output of the PS_3 code in the PS1.ipynb

## 4. Add or Double

When the random integer x is: 74, the smallest number of moves is: 16

## 5. Dynamic programming

### 5.1

The detailed results are provided in the output of the PS_5 5.1 code in the PS1.ipynb

## 5.2

For integer 1, The number of the solution is 11
For integer 2, The number of the solution is 26
For integer 3, The number of the solution is 11
For integer 4, The number of the solution is 18
For integer 5, The number of the solution is 8
For integer 6, The number of the solution is 21
For integer 7, The number of the solution is 12
For integer 8, The number of the solution is 17
For integer 9, The number of the solution is 8
For integer 10, The number of the solution is 22
For integer 11, The number of the solution is 12
For integer 12, The number of the solution is 21
For integer 13, The number of the solution is 11
For integer 14, The number of the solution is 16
For integer 15, The number of the solution is 15
For integer 16, The number of the solution is 20
For integer 17, The number of the solution is 8
For integer 18, The number of the solution is 17
For integer 19, The number of the solution is 11
For integer 20, The number of the solution is 20
For integer 21, The number of the solution is 15
For integer 22, The number of the solution is 16
For integer 23, The number of the solution is 11
For integer 24, The number of the solution is 23
For integer 25, The number of the solution is 18
For integer 26, The number of the solution is 13
For integer 27, The number of the solution is 14
For integer 28, The number of the solution is 21
For integer 29, The number of the solution is 15
For integer 30, The number of the solution is 19
For integer 31, The number of the solution is 17
For integer 32, The number of the solution is 14
For integer 33, The number of the solution is 19
For integer 34, The number of the solution is 19
For integer 35, The number of the solution is 7
For integer 36, The number of the solution is 14
For integer 37, The number of the solution is 19
For integer 38, The number of the solution is 19
For integer 39, The number of the solution is 17
For integer 40, The number of the solution is 18
For integer 41, The number of the solution is 16
For integer 42, The number of the solution is 17
For integer 43, The number of the solution is 18
For integer 44, The number of the solution is 10
For integer 45, The number of the solution is 15
For integer 46, The number of the solution is 26

For integer 47, The number of the solution is 18
For integer 48, The number of the solution is 15
For integer 49, The number of the solution is 16
For integer 50, The number of the solution is 12
For integer 51, The number of the solution is 17
For integer 52, The number of the solution is 19
For integer 53, The number of the solution is 9
For integer 54, The number of the solution is 17
For integer 55, The number of the solution is 21
For integer 56, The number of the solution is 16
For integer 57, The number of the solution is 13
For integer 58, The number of the solution is 14
For integer 59, The number of the solution is 16
For integer 60, The number of the solution is 17
For integer 61, The number of the solution is 17
For integer 62, The number of the solution is 11
For integer 63, The number of the solution is 13
For integer 64, The number of the solution is 22
For integer 65, The number of the solution is 14
For integer 66, The number of the solution is 13
For integer 67, The number of the solution is 15
For integer 68, The number of the solution is 15
For integer 69, The number of the solution is 15
For integer 70, The number of the solution is 17
For integer 71, The number of the solution is 7
For integer 72, The number of the solution is 14
For integer 73, The number of the solution is 17
For integer 74, The number of the solution is 15
For integer 75, The number of the solution is 12
For integer 76, The number of the solution is 13
For integer 77, The number of the solution is 14
For integer 78, The number of the solution is 14
For integer 79, The number of the solution is 14
For integer 80, The number of the solution is 10
For integer 81, The number of the solution is 9
For integer 82, The number of the solution is 19
For integer 83, The number of the solution is 12
For integer 84, The number of the solution is 13
For integer 85, The number of the solution is 13
For integer 86, The number of the solution is 12
For integer 87, The number of the solution is 11
For integer 88, The number of the solution is 12
For integer 89, The number of the solution is 6
For integer 90, The number of the solution is 12
For integer 91, The number of the solution is 14
For integer 92, The number of the solution is 16
For integer 93, The number of the solution is 13
For integer 94, The number of the solution is 11
For integer 95, The number of the solution is 11

For integer 96, The number of the solution is 10
For integer 97, The number of the solution is 11
For integer 98, The number of the solution is 7
For integer 99, The number of the solution is 9
For integer 100, The number of the solution is 17
Therefore, the plot can be obtained as below:

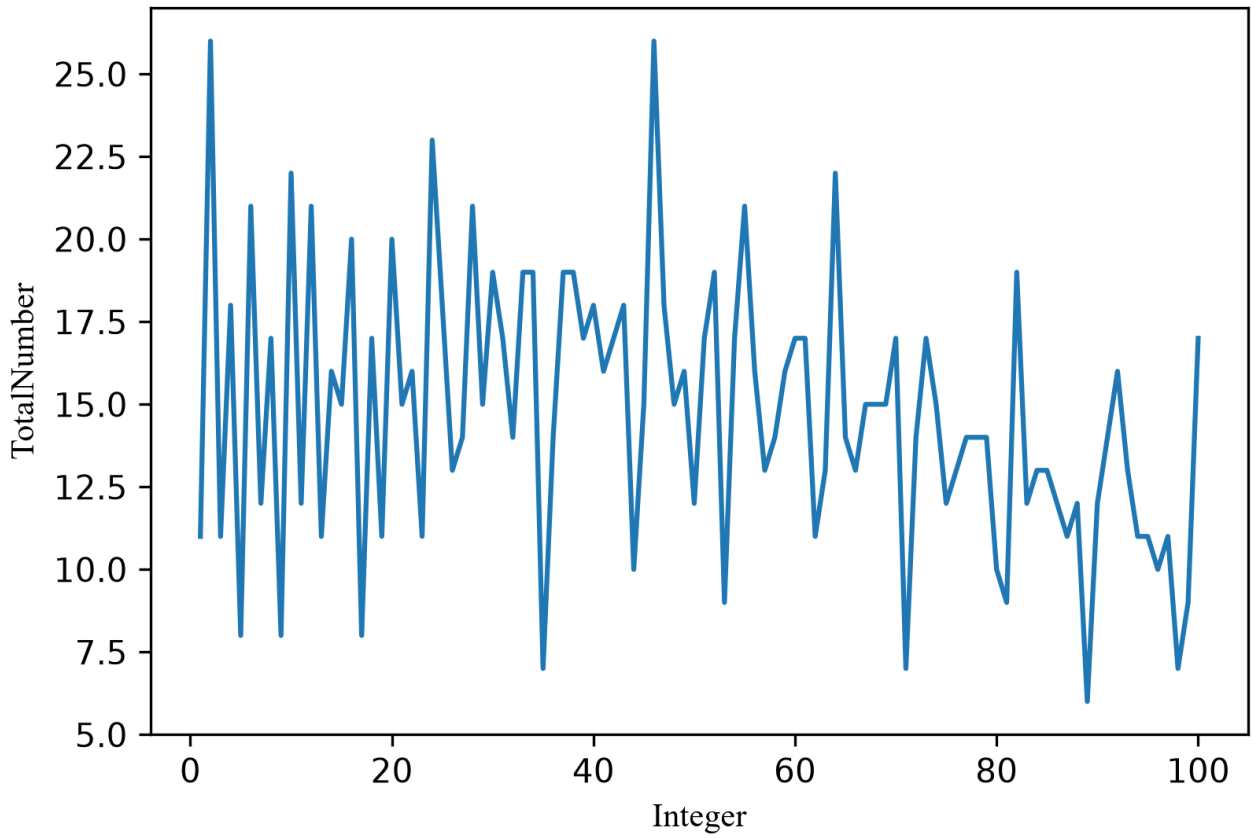



Figure 1

Number 89 yields the minimum of Total_solutions, who has 6 solutions
Number 2 yields the maximum of Total_solutions, who has 26 solutions